

Revised Generation Interconnection System Impact Study Report

For Queue Project Q18-0931

Longdon 138 kV
November 29, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0931. The Interconnection Customer proposes a 68.5 MW solar photovoltaic facility interconnecting at the Longdon 138 kV substation in the Kentucky area. The request was received on January 26, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 291 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,106,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Wec / Mirabel 138.0 kV, the Wec / Wellington 138.0 kV loads to 203.3% of its 243 MVA rating. Project DFAX: 28.3%.

For the loss of Davisville / Wellington 138.0 kV, the Davisville / Mirabel 138.0 kV L loads to 197.1% of its 217 MVA rating. Project DFAX: 18.5%.

Wec / Mirabel 138.0 kV line: 195.7% loading (222 MVA) following the outage of the Wellington / Davisville 138.0 kV; DFAX = 18.5%.

For the loss of Spring Creek Rd / Harris Substation 138.0 kV, the Turkey Creek / Tom Watkins 138.0 kV loads to 148.2% of its 228 MVA rating. Project DFAX: 7.3%.

For the loss of Spring Crk Road / Harris 138.0 kV, the Amity / Consolidated Coal 138.0 kV loads to 138.5% of its 278 MVA rating. Project DFAX: 11.3%.

The Amity / Massapequa 138.0 kV Line is overloaded to 135.6% of its 275 MVA emergency rating for the loss of the Spring Crk Rd / Harris 138.0 kV. The project contributes 11.3% DFAX (7.8 MW) to this overload.

For the loss of Wansley Combined Cycle / Massapequa 138.0 kV, the Harris / Spring Crk Rd 138.0 kV L loads to 135.0% of its 400 MVA rating. Project DFAX: 15.4%.

For the loss of Spring Creek Rd / Harris 138.0 kV, the Meeker / Turkey Creek 138.0 kV loads to 132.4% of its 260 MVA rating. Project DFAX: 7.3%.

Barnabie Cable Sta / Longdon 138.0 kV: 131.3% loading (217 MVA) following the outage of the Moss Bluff Substation / Barnabie Cable Sta 138.0 kV L; DFAX = 68.9%.

Bulan / Hazard 138.0 kV Ckt 1: 130.5% loading (374 MVA) following the outage of the Hazard / Bulan 138.0 kV; DFAX = 10.3%.

For the loss of Spring Crk Rd Substation / Harris Substation 138.0 kV line, the Meeker / Sidney Coal 138.0 kV line loads to 130.3% of its 275 MVA rating. Project DFAX: 7.3%.

The Kaeah 1 / Wellington 138.0 kV is overloaded to 128.8% of its 237 MVA emergency rating for the loss of the Davisville / N Appleton 138.0 kV. The project contributes 22.0% DFAX (15.1 MW) to this overload.

For the loss of Davisville / N Appleton 138.0 kV, the Ayers Island Substation / Kaweah 1 138.0 kV line loads to 127.6% of its 236 MVA rating. Project DFAX: 22.0%.

For the loss of Massapequa / Wansley Combined Cycle 138.0 kV L, the Hazard / Harris Substation 138.0 kV loads to 126.3% of its 299 MVA rating. Project DFAX: 10.2%.

Admin / Spring Creek Road 138.0 kV: 123.2% loading (227 MVA) following the outage of the Atoka / Spring Crk Rd 138.0 kV; DFAX = 7.1%.

The Bulan / Hazard 138.0 kV Line 2 is overloaded to 122.9% of its 398 MVA emergency rating for the loss of the Bulan / Hazard 138.0 kV Circuit 1. The project contributes 10.3% DFAX (7.1 MW) to this overload.

Granite City Works Substation / Tule 138.0 kV Line: 122.7% loading (196 MVA) following the outage of the N Appleton / Davisville 138.0 kV Line; DFAX = 7.0%.

For the loss of Hazard / Harris 138.0 kV, the Glen Carter / Starkey Gulch 138.0 kV loads to 122.3% of its 216 MVA rating. Project DFAX: 6.5%.

The Admin / Lank 138.0 kV L is overloaded to 121.2% of its 225 MVA emergency rating for the loss of the Atoka / Spring Crk Rd 138.0 kV. The project contributes 7.1% DFAX (4.8 MW) to this overload.

Central Campus Unm / Henry Crk 138.0 kV: 109.7% loading (177 MVA) following the outage of the Albion Rd / Platte City 138.0 kV; DFAX = 12.9%.

For the loss of Hazard / Harris 138.0 kV line, the Rinway / Starkey Gulch 138.0 kV loads to 109.1% of its 231 MVA rating. Project DFAX: 6.5%.

The Wansley Combined Cycle / Massapequa 138.0 kV line is overloaded to 108.9% of its 437 MVA emergency rating for the loss of the Spring Crk Rd / Harris 138.0 kV. The project contributes 14.7% DFAX (10.1 MW) to this overload.

Harris / Reinway 138.0 kV: 106.2% loading (233 MVA) following the outage of the Harris / Hazard 138.0 kV; DFAX = 6.5%.

For the loss of Wansley Combined Cycle / Massapequa 138.0 kV, the Spring Creek Rd / Atoka 138.0 kV line loads to 105.1% of its 359 MVA rating. Project DFAX: 11.8%.

The Albion Rd / Central Campus Unm 138.0 kV is overloaded to 104.8% of its 195 MVA emergency rating for the loss of the Platte City / Albion Road 138.0 kV. The project contributes 12.9% DFAX (8.9 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Wec / Wellington 138.0 kV	Rebuild line with higher-capacity conductor	\$23,530,000	6% (\$1,415,000)
Mirabel / Davisville 138.0 kV	Rebuild line with higher-capacity conductor	\$29,590,000	5% (\$1,413,000)
Wec / Mirabel 138.0 kV line	Rebuild line with higher-capacity conductor	\$4,569,000	5% (\$218,000)
Turkey Creek / Tom Watkins 138.0 kV	Rebuild line with higher-capacity conductor	\$9,684,000	4% (\$437,000)
Amity / Consolidated Coal 138.0 kV	Rebuild line with higher-capacity conductor	\$14,245,000	5% (\$690,000)
Amity / Massapequa 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,232,000	5% (\$157,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Spring Crk Road / Harris 138.0 kV	Rebuild line with higher-capacity conductor	\$3,266,000	4% (\$132,000)
Meeker / Turkey Crk 138.0 kV	Rebuild line with higher-capacity conductor	\$862,000	4% (\$39,000)
Longdon / Barnabie Cable Sta 138.0 kV line	Rebuild line with higher-capacity conductor	\$13,147,000	19% (\$2,471,000)
Bulan / Hazard 138.0 kV	Rebuild line with higher-capacity conductor	\$2,817,000	5% (\$152,000)
Meekr / Sidney Coal 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,275,000	4% (\$283,000)
Kaeah 1 / Wellington 138.0 kV	Rebuild line with higher-capacity conductor	\$4,981,000	6% (\$288,000)
Ayers Island / Kaweah 1 138.0 kV	Rebuild line with higher-capacity conductor	\$29,583,000	6% (\$1,711,000)
Hazard / Harris Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$39,514,000	5% (\$1,975,000)
Admin / Spring Creek Road 138.0 kV	Rebuild line with higher-capacity conductor	\$12,593,000	5% (\$634,000)
Bulan / Hazard 138.0 kV Line 2	Rebuild line with higher-capacity conductor	\$2,151,000	5% (\$116,000)
Granite City Works Substation / Tule 138.0 kV Line	Rebuild line with higher-capacity conductor	\$38,814,000	3% (\$1,318,000)
Glen Carter / Starkey Gulch 138.0 kV	Rebuild line with higher-capacity conductor	\$14,852,000	5% (\$710,000)
Lank / Admin 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,683,000	5% (\$185,000)
Central Campus Unm / Henry Crk 138.0 kV	Reconductor line	\$4,051,000	4% (\$150,000)
Sarkey Gulch / Reinway 138.0 kV	Reconductor line	\$3,086,000	5% (\$148,000)
Wansley Combined Cycle / Massapequa 138.0 kV Line	Reconductor line	\$2,106,000	4% (\$89,000)
Harris / Reinway 138.0 kV	Reconductor line	\$3,272,000	5% (\$156,000)
Atoka / Spring Creek Road 138.0 kV	Reconductor line	\$4,114,000	4% (\$168,000)
Albion Rd / Central Campus Unm 138.0 kV	Reconductor line	\$1,391,000	4% (\$51,000)
	Total allocated to project		\$15,106,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.